J. Int. Financ. Markets Inst. Money 75 (2021) 101304

Contents lists available at ScienceDirect

Journal of International Financial Markets, Institutions & Money

journal homepage: www.elsevier.com/locate/intfin

Real Output and Cross-Currency Basis Swap Spreads: Evidence from the Eurozone

Oyakhilome Ibhagui

Baum Tenpers Finance Research Group, Nigeria

ARTICLE INFO

Article history:
Received 6 March 2019
Accepted 27 January 2021
Available online 2 May 2021

JEL Classifications:
E4
E5
E6
F3
G1
G2

Keywords:
Eurozone
Exchange rates
Money supply
Real output and euro-dollar cross-currency basis

ABSTRACT

Can higher real output tighten covered interest parity (CIP) deviations and lessen arbitrage profits in the eurozone fixed income market for dollar-based global investors? We address the question of determining the long-run links between selected eurozone macroeconomic factors and the long-end of the EUR/USD cross-currency basis swap spread. In our stylized model, a rise in eurozone relative real output tightens the cross-currency basis and hence lowers available arbitrage profits for USD investors looking to execute arbitrage trades whereas an increase in relative money supply together with euro depreciation widens the basis and increases potential arbitrage profits. The magnitude of the effect of relative real output dominates the magnitude of the individual effects of relative money supply and exchange rate. Our empirical results are fully consistent with the stylized model.

1. Introduction

A major shift in the global financial markets over the last decade - one which has added an important variable to the list of carefully monitored indicators in the financial markets - is the persistent breakdown of covered interest parity (CIP). At its core, the CIP is a no arbitrage condition. In simple terms,[1] it states that, given two currencies \$ and €, the income realized when a \$ holder invests in a € fixed income security and exchanges the € proceeds to \$ must be equivalent to the income generated

E-mail address: wallace@aims.ac.za

[1] How a pure arbitrage profit from CIP deviations - manifested as a negative cross-currency basis between the euro and U.S. dollars – can be made. We start by assuming the arbitrageur has no dollar in hand. To arbitrage in the euro market, the U.S. dollar arbitrageur borrows one \$1 at the interest rate i and converts the \$1 into euro at a rate of \$1 to S euro. This gives S euro. The arbitrageur then approaches the euro zone financial market, lends out the euro at the prevailing euro market interest rate i*, and then signs a forward contract on the same date to convert the euro proceeds to the U.S dollar at a future date when the deal expires, i.e. at maturity or end of investment horizon. At maturity, which is at the end of the investment horizon n, the arbitrageur receives $Se^{ni^*} \simeq S(1+i^*)^n$ in euros, and converts this sum into $Se^{ni^*}/F = S(1+i^*)^n/F$ U.S. dollars due to the forward contract F. At the end of the investment horizon, the arbitrageur's \$1 debt in the U.S. dollars together with the accrued interest would be e^{ni}. The arbitrageur repays this debt in U.S. dollars from Se^{ni^*}/F which is the sum in U.S. dollars realized from investing in the euro market. The arbitrageur is then left with a profit of $(Se^{ni^*}/F - e^{ni})$ which is equal to the absolute value of the euro cross-currency basis. Thus, the more negative is the euro cross-currency basis, the greater the arbitrage profit that new arbitrage positions can generate for the arbitrageur. In summary, the arbitrageur has made a profit from owning. The arbitrageur starts out by borrowing dollars, exchanging to euro, going long in the euro asset (and shorting the dollar), with the euro cash flow fully hedged back to dollars using forwards.

https://doi.org/10.1016/j.intfin.2021.101304

had the $ holder invested in a similar $ security domestically. This also implies that when a euro entity issues in $ and exchanges the cost of the $ funds raised to €, this cost must be equal to that incurred had the euro entity issued a similar product domestically in €. Otherwise, it would be possible to raise cheaper € funds via issuing in $ or to obtain a higher income in $ via investing in € bonds, with currency risk fully hedged. The CIP breakdown became most visible in the last crises and has since persisted – offering the possibility to access cheaper funds or make arbitrage profits than would have been possible had the CIP remained largely unbroken. The CIP breakdown is encoded in what has now been known as the cross-currency basis swap spread. If CIP were unbroken, this spread would be zero. Thus, the non-zero nature of the spread captures the magnitude or extent of CIP breakdown. This variable also indicates the level of stress in the financial market. For the euro basis, it is highly negative across maturities, including at the low volatility and less erratic long end of the basis curve which is the focus of this paper (see Tables 1–4).

Much of the recent studies on cross-currency basis swap spreads have been skewed towards explaining why risk-free arbitrage still exists and why the spreads have not fizzled out, what the spreads imply about counterparty risks in currencies, how they relate to the general demand for, and supply of, dollar-based assets and how post-crisis regulations - which impose balance sheet constraints that limit the size of exposures that banks and non-banks can take - have impacted the basis spreads. One of the major findings in the literature is that the CIP breakdown persists because banks, and other financial institutions that depend on them, are constrained, by increased regulations post crisis, from exploiting the opportunities which the CIP breakdown affords. As a result, the opportunities remain, because they have not been fully exploited, and the euro cross-currency basis remains deeply negative across maturities. But how do macroeconomic factors contribute to the persistence of the deeply cross-currency basis in the eurozone? The role of macroeconomic factors in the persistence of the euro cross-currency basis has remained largely ignored in the literature. Since financial markets necessarily rely on benign macroeconomic conditions, it makes sense to think that relevant information on cross-currency basis swap spreads should be contained in some macroeconomic factors. The focus of this paper, therefore, is on what we can learn about the euro cross-currency basis swap spread from macroeconomic factors in the euro area.

Another motivation for this study is that the limited existing literature on CIP breakdown has focused almost exclusively on the strength of the dollar exchange rate as a major factor responsible for driving the cross-currency basis wider, in line with the findings by Avdjiev et al. (2018). Although this is true and is confirmed in our study, we believe this does not tell the full story. First, whilst it is true that an increase in dollar strength could reduce dollar flows to other countries and lessen dollar-lending activities – as the fear of potential default increases given that a stronger dollar burdens the ability of borrowers or counterparties to meet obligations, which would slow cross-border dollar flows and reduce appetite for non-dollar assets – it is also important to note that other factors, such as a more favorable economic performance, exist that could raise cross border dollar flows even amid an increasing dollar strength. As such, the dollar will only be dominant and exert the most influence on the cross-currency basis swap spreads when other indicators of financial and economic health are much less benign. Second, an exclusive focus on the dollar neglects the potentially substantial impact of broader economic drivers on the cross-currency basis swap spreads, an effect already emphasized by the low coefficient of determination for the dollar and other non-macroeconomic factors as the main driver of the basis in existing studies. We seek to address these highlighted gaps in the literature by emphasizing the role played not only by the dollar exchange rate but also by macroeconomic factors – such as the real and monetary aspects of the economy – in determining behavior of the cross-currency basis swap spreads.

To understand the impact of eurozone macroeconomic factors on the euro cross-currency basis, we construct a stylized model relating the euro cross-currency basis with standard macroeconomic factors – exchange rate, money supply and real output representing relative currency prices, monetary policy and performance of the real economy, respectively. We model the cross-currency basis as a deviation from CIP, in line with its standard definition. Our model suggests a tightening of the euro cross-currency basis in the long run when real output increases in the eurozone relative to the U.S., but a widening of the basis following a rise in eurozone money supply relative to the U.S. and a depreciation of the exchange rate. We take these predictions to euro data samples and directly investigate their implications for the euro cross-currency basis. We do this to ensure that the theoretical results are not a mere coincidence but adequately fit and are consistent with the euro data. We first examine the non-stationarity and cointegration property of the euro data. Our results show that the macroeconomic variables and euro cross-currency basis are cointegrated, implying the existence of an estimable long-run relationship in levels. To estimate the long-run relationship, we employ appropriate estimators for cointegrated time series such as dynamic OLS, fully modified OLS and canonical cointegrating regression.

After estimating the coefficients on the macroeconomic variables, we find that our results are consistent with those implied by the model for the period considered in that they show an economically and statistically significant long run relationship between macroeconomic factors and cross-currency basis swap spreads in the eurozone. The key message from the stylized model and eurozone data is that macroeconomic factors have played a significant role in the behavior of the cross-currency basis in the eurozone. In particular, we find that real output, money supply and the exchange rate are linked with the cross-currency basis in the long run; the linkage is such that (i) a rise in euro area output relative to US real output tightens the basis, (ii) an increase in euro area money supply relative to US money supply widens the basis and (iii) a depreciation of the EURUSD exchange rate also widens the basis. We find that these results are generally true across all estimators employed. Thus, deviations from CIP in the eurozone turn on the exchange rate and the scale of money supply in the eurozone. When eurozone relative money supply rises and the dollar strengthens relative to the euro, deviations from CIP become larger and the cross-currency basis tends to widen.

Our point estimates across all estimators considered indicate that, other things being equal, a one percent increase in eurozone relative real output tightens the 10-year euro cross-currency basis by 3.43–4.56 units per quarter, while a one percent depreciation of the exchange rate and increase in euro area relative money supply widen the 10-year euro cross-currency basis by 1.47–3.22 units and 0.97–2.15 units, respectively. This indicates that most of the tightening in the long end of the euro basis has been driven by improvements in real output in the eurozone. Not only is this important, it is able to dilute the widening pressure from currency depreciation and money supply increases in the eurozone as shown in the point estimates. Thus, sustainably higher real output in the eurozone can potentially tighten CIP deviations associated with the euro and thus lessen potential arbitrage profits available to dollar-based investors interested in the eurozone fixed income market. It should be noted that our framework is for long-term relationships as we have established long-run equilibrium relationships that are based on the empirical evidence of cointegration obtained for the levels variables in the period analyzed. Thus, we depart from studies that examine short-term relationships using changes of variables.

We also investigate the dynamic impact of the error correction term on the changes of each of the variables in the cointegrated vector to get a sense of the adjustment back to our estimated equilibrium following a temporary departure. We find that the exchange rate and, to a less significant extent, the euro cross-currency basis, adjusts to restore the estimated long-run relation. With respect to the other macroeconomic variables, the error correction variable has either weak power for their dynamics or yields a positive sign unsupportive of convergence.

We have focused on the euro area for several reasons. First, there has been a significant increase of international bond issuance in euro since the launch of the European monetary union, Hale and Spiegel (2012). Also, Avdjiev et al. (2018) find evidence that the euro has started to exhibit characteristics of a global funding currency. This makes a study examining the link between eurozone macroeconomic factors and cross-currency basis very important especially given that proceeds from euro issues are often of intermediate importance since the final currency of relevance for many these issues, especially issues by large multilateral development institutions, is often in a different currency from the euro (i.e. they are swapped to the US dollars). Second, the euro cross-currency basis swap market is the most active among peers, accounting for 46% of volumes (see Barnes (2017)). So, the euro basis is the largest cross-currency swap market. Moreover, the EUR/USD is a very liquid currency pair and is the most traded foreign exchange pair among the G10 currencies (see Gargano et al. (2018)), accounting for over 20% of the total volume of transactions in the forex market, thus providing adequate liquidity and required variability in the data for analysis. Third, the eurozone is the largest and best-known monetary union which affords the region with macroeconomic scale and size, a broader-based and synchronized monetary policy, and increases the internationalization of the euro in global payments and foreign exchange settlements.

To provide a graphical illustration of what turns out to support our stylized model and empirical finding, Figs. 1a–1c below plots the cross-currency (in blue) paired with the relative real output (ash), relative money supply (purple) and exchange rate (black) over time. A cursory look at the graphs suggest that when the ash line goes down, which means the eurozone real output underperforms the US output, the cross-currency basis (blue line) tends to widen. We also see that when the purple line increases, which means that the eurozone money supply increases relative to the US money supply, the cross-currency basis tends to widen. In fact, as it can be seen, the cross-currency basis, by and large, is often the mirror image of the relative money supply. Lastly, the plot of the EUR/USD exchange rate are shown below in black. When the lines go up, the dollar strengthens (or the euro weakens) and when they go down, the euro strengthens, or the dollar weakens. Again, as with money supply, we see that the euro cross-currency basis is the mirror image of the euro weakness or dollar strength. When the euro weakens (or the dollar strengthens), the cross-currency basis spreads widen.

The rest of the paper is organized as follows. Section 2 provides a review of the related literature and comparison with other empirical studies. Section 3 reports some stylized facts about the euro basis while Section 4 presents the baseline model framework which forms the foundation of subsequent empirical analysis. Section 5 explains the data, some preliminary commentary, methodology and empirical results while Section 6 analyzes the convergence to long run via error-correction analysis. The last section concludes.

2. Relation to other studies

Although to our knowledge, there are limited studies that have addressed the links between the identified macroeconomic variables and the cross-currency basis in a major monetary union such as the eurozone, we would attempt to make a comparison of our work with a few other related studies available in the literature and thus comfortably place our contributions in the literature.

Using a collection of G10 countries, Avdjiev et al. (2018) estimated a panel regression of the cross-currency basis on the dollar exchange rate, with the variables expressed as changes in levels over the sample period. They obtained a strong evidence of a wider basis when the dollar appreciates, with a significantly negative coefficient across most currencies. In this paper, we have estimated the long-run links not only between exchange rate and cross-currency basis for a longer span but also, we have done this in the presence of other macroeconomic drivers of the basis. This enables us to account, as suggested by the stylized model, for the effects of relative money supply and relative real output. This is important, especially in view of the links between these variables and the basis, the strong and significant effect of relative money supply and relative real output on the basis at the long end, and the fact that exchange rate is not necessarily the only important driver of the basis in the long run. This path we have taken is relevant as it enables us to obtain the effect of each variable on the cross-currency

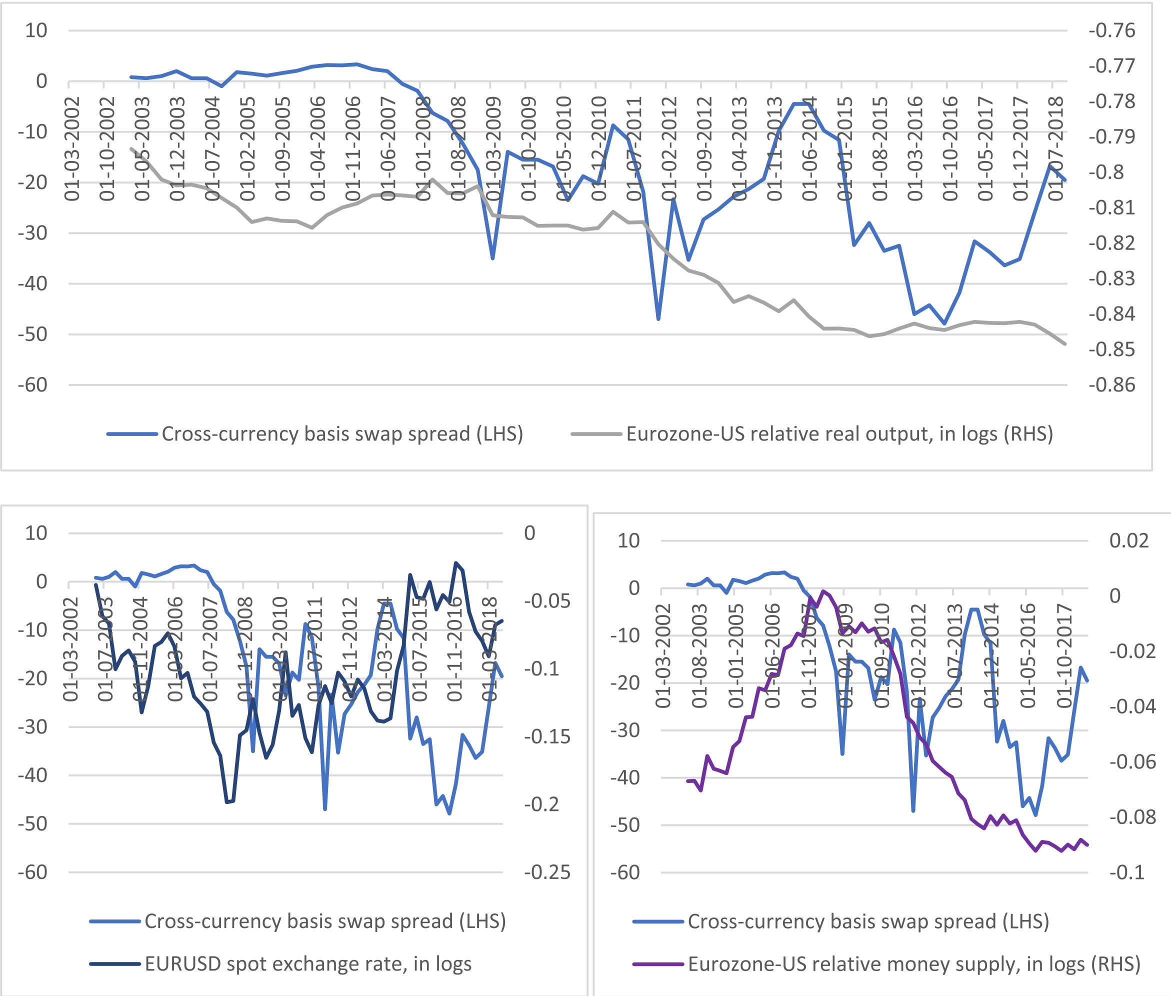


Fig. 1a. 10-year cross-currency basis paired with relative output, money supply, spot and forward exchange rate.

basis whilst accounting for the other macroeconomic variables suggested in the stylized model. We found the effect of relative real output to be much stronger, and the effect of relative money supply and exchange rate to be much weaker in our framework.

With respect to long-run cointegrating relation, Baran and Whitzany (2017) estimate the long-run drivers of the euro cross-currency basis, including in their cointegrating vector variables such as exchange rate, measures of credit risk, volatility index and rates. They find that the cross-currency basis tends to be tighter in the long run when the euro strengthens relative to the dollar. However, their study excludes the macroeconomic variables we have considered in this paper and are difficult to interpret comparatively given the discrepancy in the units of measure for the variables included in their cointegrating vectors. Du et al. (2018) estimate the effect of regulation on cross-currency basis and find that the more stringent regulation that followed the crisis explains why the cross-currency basis has persisted, and CIP deviations have not been arbitraged away. The key issue in these studies is the neglect of the effects of other plausible macroeconomic factors on the cross-currency basis. Moreover, ignoring these factors have possibly been responsible for the often-low explanatory power of the drivers of the basis advanced so far in the literature.

In terms of how our paper places or fits into the largely expanding literature on deviations from CIP, we note that our study is most related to the growing strand of literature that examines the macroeconomic drivers of cross-currency basis swap spreads that surfaced since deviations from CIP became persistently nonvanishing. For example, since the paper highlights the nexus between macroeconomic variables and these nonvanishing deviations, then it is related to studies such as Cerutti et al. (2019) who argue for macro-financial variables as forces that have become significant drivers of deviations from CIP, Costa and Marcal (2019) who show that CIP deviations respond to variations in traditional macro variables – public debt, foreign reserves, inflation, and trade openness, Ibhagui (2020) who demonstrate that higher inflation differential drives CIP deviations wider, and Liao and Zhang (2020) who show that in times of economic stress, increased exchange rate hedging

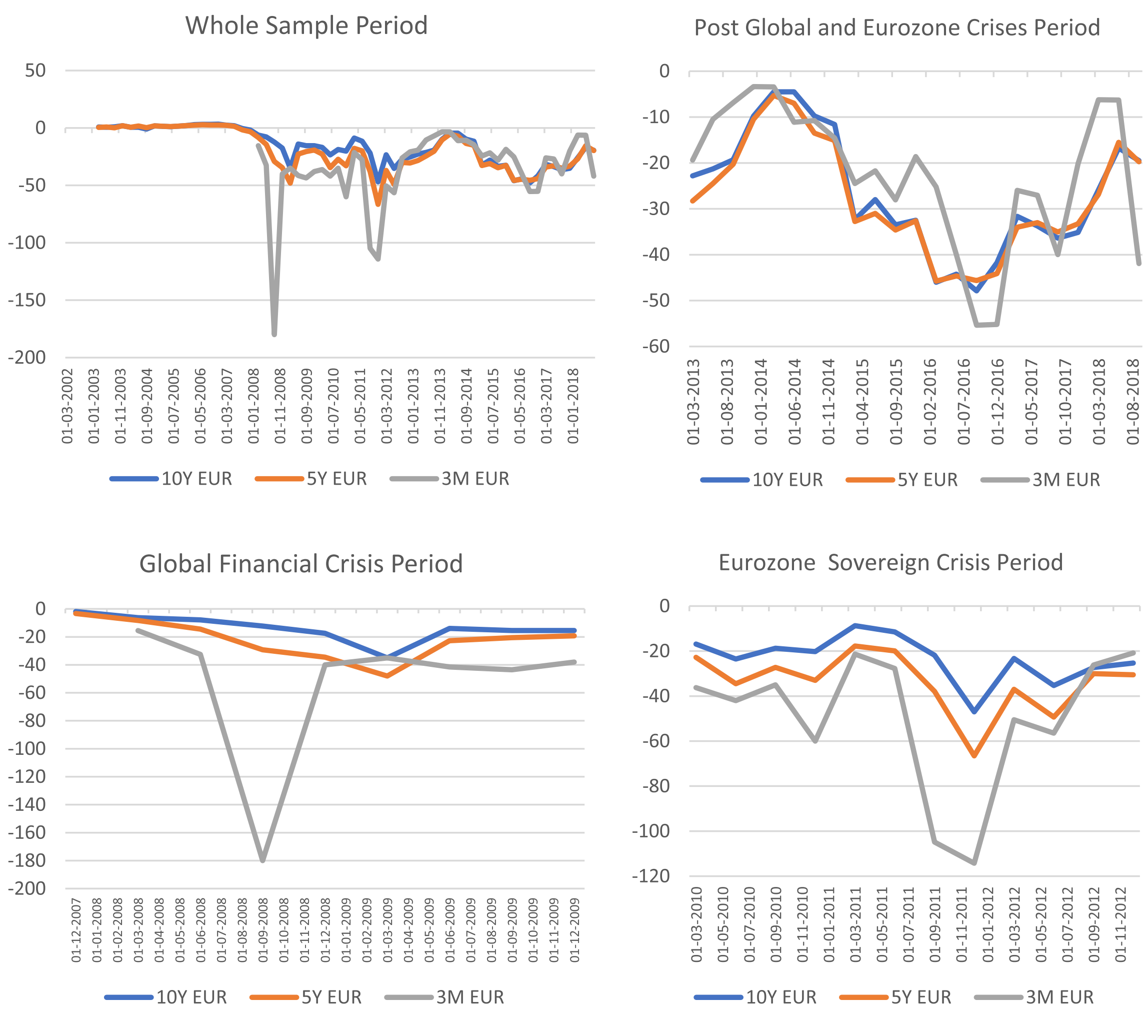


Fig 1b. Evolution of the euro cross-currency basis swap spreads. Source: Author's calculation, Bloomberg.

activities explain some of the widening and contraction in CIP deviations observed in countries with positive and negative net foreign assets position, respectively.

As we focus on macroeconomic factors and Libor-based CIP deviations, our work differs, for example, from Du et al. (2018b) and Jiang et al. (2018, 2020). Du et al. (2018b) seminal work documents the negativity of U.S. Treasury basis. Jiang et al. (2018) provide evidence that demand for US dollar safe assets drives yields for these assets lower and hence Treasury basis wider (i.e. more negative), which cause substantial dollar appreciation. Jiang et al. (2020) establish that persistent negative Treasury basis reflects the sacrifice that agents are willing to make in order to hold US dollar assets by way of the high premium they place on cash dollar safe assets, which drives yields on these assets lower, i.e. the convenience yield on dollar safe assets. Also, our finding that cross-currency basis can be driven by macroeconomic forces is distinct from the perspective of intermediary constraints as highlighted in recent studies, for instance Du et al. (2019) and Correa et al. (2020).

In terms of the choice of basis analyzed, our focus on the EUR/USD basis differs from Costa and Marcal (2019), Coskun and Ibhagui (2020), and Hartley (2020), all of which are related to our paper in scope but distinct in coverage as they focus solely on deviations from CIP in EM cross-currency basis markets. For instance, Coskun and Ibhagui (2020) show that economic-enhancing technology shocks induce a positive (tightening), but sometimes mixed, response from EM CIP deviations, consistent with their model of domestic technology shocks as the dominant source of gains in economic output. Hartley (2020) provides a seminal exploration of covered interest rate parity violations by non-dollar denominated external EM bonds relative to their dollar-denominated external EM bonds. Among others, they find such violations to have created opportunities for EM policymakers to create fiscal savings by issuing external sovereign debt in dollars and swapping proceeds to non-dollar currencies on a hedged basis.

Importantly, in all parts of this paper, the basis we refer to and analyze is the Libor-based cross-currency basis swap spread as in Du et al. (2018a), Avdjiev et al. (2018), Ibhagui (2020), among others. This is the basis defined as direct dollar

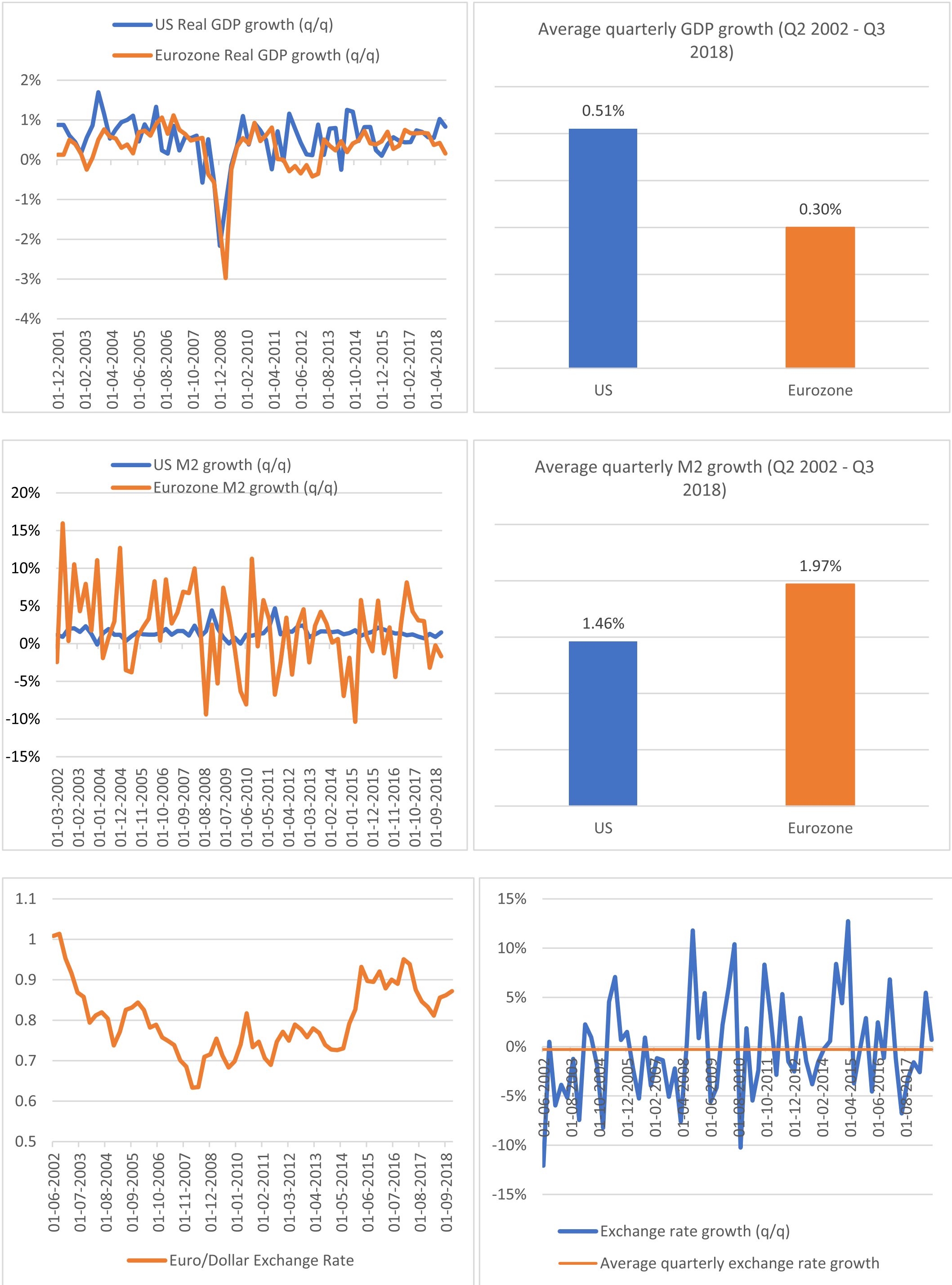


Fig. 1c. Graphical view of the explanatory macroeconomic variables. Source: Author computation, FRED and Bloomberg.

interest rate cost in the cash market less indirect/synthetic dollar cost in the swap market obtained by swapping other currencies to the dollar. It is based on the Libor benchmark, not on Treasury yield (as in Treasury basis, see Du et al., 2018b) or corporate yield (as in corporate basis, see Liao, 2020). In our framework, the Libor-based CIP deviation measures the basis using the Libor benchmark of the US dollar and foreign currency Libor benchmark as the interest rates, whereas Treasury-based and corporate-based CIP deviations measure the basis using Treasury and corporate bond yields respectively as the interest rate benchmarks. Nonetheless, Jiang et al. (2020) have demonstrated that the three bases – Libor, Treasury and corporate – typically move together, with the Treasury basis often wider (more negative), while (Liao, 2020) notes that the Treasury and corporate basis can be decomposed into a Libor CIP deviation component and a credit risk component, which supports the closely knit movement often observed in the three bases.

Our paper is the first to directly study the long-run relationship between the euro cross-currency basis and standard macroeconomic variables in the eurozone. The only other paper that we are aware of, that specifically studies the long-run drivers of the euro basis, is Baran and Whitzany (2017). However, their framework is purely empirical, focuses on the short to mid end of the euro basis and does not provide a theoretical motivation. We have presented, instead, a new empirical evidence motivated by a simple theoretical framework which enables us to directly test the implications of our hypothesis without losing much degree of freedom in an environment of limited data, as that would be the case if we uncoordinatedly handpicked regressors as potential drivers of the basis. In a recent work, Ibhagui (2020) derives a model based on the standard Cagan-type monetary framework. This paper, which focuses on the eurozone, rather than on a panel of countries like Ibhagui (2020), draws from that model as a benchmark to investigate the significance of the identified macroeconomic factors for the euro basis in the long run. Thus, our results are specific to the eurozone and we make no claim that they extend to other countries or currencies or even periods outside those we have considered, given the heterogeneity among countries, even among those with developed financial markets.

Finally, this paper has investigated long-run relations between the basis and macroeconomic factors, with a focus on the eurozone. Our results suggest directions for future research. Thus, our next line of action would be to extend this country specific research to include other G10 currencies and the broader emerging markets even as we also seek to analyze the short run dynamic linkages between the basis and potential macroeconomic drivers. In countries for which there is support for our model, it would be interesting to compare the similarity of the support to the eurozone and to examine the dynamics of the adjustment process to the long-run equilibrium relation implied by the model by studying metrics such as the panel impulse responses and variance decompositions for the cross-currency basis and macroeconomic fundamentals in these countries. It would also be interesting to undertake a research to build a dynamic stochastic general equilibrium (DSGE) model that can replicate the dynamics of these variables and extend this line of research to emerging market economies. However, the task could be complicated by patchy data as the derivatives markets of some of these EM economies are much less developed. This paper has taken the U.S. as the numeriare. Future research could utilize other major countries or even the eurozone as the numeriare and investigate whether our model has support as a long-run anchor for characterizing the cross-currency basis. Finally, it is plausible that the divergence of the basis from the identified underlying macroeconomic factors could exhibit nonlinearity in its reversion to mean. Thus, issues pertaining to exploring this possibility for the euro zone and other countries for which we find support for our model have major relevance for derivatives policy makers and rank high in the priority list for future research efforts in untangling the long-run macroeconomic anchors for the cross-currency basis.

3. The EUR/USD basis – stylized fact

Here, we present some stylized facts on the euro cross-currency basis swap spreads. We take a cursory look at the basis at the 3-month, 5-year and 10-year maturity buckets and across notable periods over the past decades. The graphs below summarize the quarterly behavior of the basis across the three maturities over the (i) whole sample period, (ii) global financial crisis period (2007 to 2009), (iii) eurozone crisis period (2010 to 2012), (iv) combined crisis periods (2007 to 2012) and, finally, (v) the period post crisis (2013 to 2018), respectively.

We begin with the whole sample period. In this case, across the three maturities, the 3-month euro basis is the widest, on average, and has the highest volatility. This is followed by the 5-year euro basis while the 10-year euro basis is the tightest and has the least volatility across the three maturities. The 3-month EUR basis seems to diverge significantly from the 5-year and 10-year basis, both of which appear to be well synchronized. The correlation analysis presented below confirms this observation quite well.

In the above correlations, the 5-year and 10-year basis swap spreads have the highest correlation of 0.96, compared to the correlation of 0.49 and 0.19 between the 3-month basis and each of the 5-year and 10-year basis respectively. These correlations are significantly smaller, confirming the weak harmonization between the short end of the basis (3-month) and those at the longer end (5-year and 10-year).

Turning now to the two crises – that is, the global financial crisis and the eurozone sovereign crisis – the account is largely similar. The short end of the basis curve is, on average, several times wider than the long end and has the highest volatility in both crises. The short end of the euro basis is the most volatile and widest compared to the long end - which remains tighter, on average, and has the least volatility. The observed low correlation pattern between the short and long end of the basis is also supported by the correlation values for both periods of crises. In fact, the 3-month basis behaves so differently that its correlation with the 10-year basis turns negative in the global financial crisis period, suggesting that there is some substitutability between the 3-month versus 10-year basis, which also implies that the basis swaps could serve as a hedge for one another. For basis trading, this outcome suggests the existence of a hedging opportunity for the euro basis with itself and has implications for risk management in a portfolio with exposure to cross-currency basis swaps at the short and long maturities.

Interestingly, however, the correlation between the basis, particularly between the short and long end, rises several times more in the eurozone crisis period compared to the global financial crisis period, suggesting that the association of the euro

Table 1
Summary statistics – whole sample period.

	10 EUR	*5 EUR*	*3 M EUR*
Mean	−12.88	−15.62	−35.76
Median	−7.04	−9.37	−27.75
Standard Dev	14.63	17.09	32.16
Range	51.21	69.33	176.63
Minimum	−47.87	−66.55	−180.00
Maximum	3.34	2.78	−3.36

Correlation - whole sample period

	10Y	*5Y*	*3M*
10Y	1.00		
5Y	0.96	1.00	
3M	0.19	0.49	1.00

Notes: The table provides summary statistics for the whole sample. The mean values show that there was tighter basis in the 10Y (Longer end basis) while 3M (short end basis) recorded the widest basis. Under the same period, we notice that 3M EUR is the most volatile (using the value of its standard deviation) compared to 10Y and 5Y. The highest value of 3.34 occurs in Q4 2006 for the long-end basis while the lowest value occurs in Q3 2008 for the short-end basis.

Table 2
Summary statistics - global financial crisis and Eurozone crisis periods.

	Global financial crisis			Eurozone crisis period		
	10Y	*5Y*	*3M*	*10Y*	*5Y*	*3M*
Mean	−13.96	−22.24	−53.25	−23.30	−33.86	−49.62
Median	−13.95	−20.55	−39.00	−22.57	−31.74	−39.12
Standard Dev	9.39	13.65	51.95	10.23	13.42	30.88
Range	33.12	44.70	164.50	38.28	48.80	93.41
Minimum	−35.00	−48.00	−180.00	−47.00	−66.55	−114.25
Maximum	−1.87	−3.30	−15.50	−8.71	−17.75	−20.83

Notes: The above summary represents the period of Global Financial Crisis (Q4 2007 to Q4 2009) and European Sovereign Crisis (Q1 2010 to Q4 2012).

Correlation				Correlation			
Global financial crisis				**Europe sovereign crisis**			
	10Y	*5Y*	*3M*		*10Y*	*5Y*	*3M*
10Y	1			10Y	1		
5Y	0.92	1		5Y	0.95	1	
3M	−0.083	0.22	1	3M	0.62	0.79	1

basis across maturities in the eurozone sovereign crisis period is not the same as during the global financial crisis. Nonetheless, in both instances, the previous theme remains – the correlation between the short end and long end of the euro basis is lower than the correlation between the long end.

The above pattern remains true when we look at the combined crisis period – i.e. combination of global and eurozone crisis periods, see Table 3, although the correlation between the short and long end drops even further, reflecting the relatively lower correlation in the global financial crisis period compared to eurozone crisis period.

However, some slight changes seem to have occurred after both crises. Indeed, looking at the after-crisis period, the long end of the basis now appears slightly wider than the short end. Nonetheless, the evidence that the short end has a higher volatility is unaltered as the basis continues to be more volatile in the short end than in the long end. Moreover, the correlation between the basis at the long end remains larger than the correlation between the short and long end of the basis, suggesting that the initial evidence that the correlation between the long and short end of the basis is relatively low remains true.

Despite this, it is important to note that the correlation between the short and long end has increased noticeably in the after-crisis period. In fact, the behavior of the euro basis post crisis is similar to the behavior in the eurozone crisis period.

In all, some interesting observations arise from the summary statistics. First, the correlation between the short and long end of the euro basis is relatively lower than the correlation between the long end. Second, the short end of the basis is more volatile than the long end and has a mostly wider mean value compared to the long end. Third, the behavior of the euro basis across maturities in the post crisis period is generally akin to its behavior during the eurozone crisis.

Table 3
Summary statistics - combined crisis period.

	10Y	*5Y*	*3M*
Mean	−19.29	−28.88	−51.0736
Median	−17.50	−29.25	−39.00
Standard Dev	10.73	14.43	39.36
Range	45.125	63.25	164.5
Minimum	−47.00	−66.55	−180
Maximum	−1.87	−3.30	−15.50

Notes: The above summary statistics is a combined period of Global Financial Crisis and European Sovereign Crisis (which overs Q4 2007 – Q4 2012). From the table we notice that 5Y and 10Y tend to behave alike while 3 M differs with the most volatile data and has the widest cross currency display.

Correlation – combined crisis period

	10Y	*5Y*	*3M*
10Y	1		
5Y	0.95	1	
3 M	0.23	0.44	1

Table 4
Summary statistics - after-crisis period.

	10Y	5Y	3M
Mean	−26.47	−27.52	−22.44
Standard error	2.72	2.56	3.21
Median	−28.00	−31.00	−20.15
Standard Dev	13.04	12.30	15.41
Range	43.38	40.45	52.01
Minimum	−47.88	−45.75	−55.37
Maximum	−4.50	−5.30	−3.36

Notes: The above summary statistics is for the post global crisis and eurozone crisis periods (which overs Q1 2013– Q3 2018). From the table we notice that

Correlation – After-crisis period

	10Y	*5Y*	*3M*
10Y	1		
5Y	0.98	1	
3M	0.75	0.75	1

We now move to the next section where we examine the baseline stylized model that proposes how macroeconomic factors might related with the basis at the long end. Our aim is to understand how the basis links with movements in these macroeconomic factors.

4. Baseline theoretical model

To illustrate how relative real output, money supply and exchange rate may influence the long-term euro cross-currency basis swap spread, we consider a continuously compounded n-year risk-free interest rates quoted in U.S. dollars and the euro – which is the foreign currency. Let i and i^* be the continuously compounded n-year risk-free interest rates at time t in the U.S. dollars and euro, respectively. Let S be the EURUSD spot exchange rate at time t, where S is expressed as the number of units of the euro per unit of U.S. dollars, so an increase in S is understood as an appreciation of the U.S. dollar (depreciation of the euro), i.e. the U.S. dollar is the numeraire currency. Furthermore, let F denote the n-year forward exchange rate in the same unit as the spot, at time t.

Suppose a representative agent, with investible U.S. dollars, seeks to exploit the markets to optimize her financial outcomes. For ease of exposition, let the investible dollars available to the agent be $1. Two outcomes are possible. First, at time t, she could invest the $1 domestically and generate a total amount of e^{ni} U. S. dollars n years from now. Second, she could exchange her $1 for the euro at the spot exchange rate S, enter a swap contract agreement and then invest the S euro in the eurozone market. At the end of the n years, she would own a total of $e^{ni^*} S$ euros. When she swaps the amount realized in euro back to U. S. dollars at the forward rate F previously agreed at time t, this amounts to $e^{ni^*} S/F$ in U. S. dollar terms. This is called synthetic US dollars.

If both the U.S. and euro securities are risk-free and forward contract bears negligible counterparty risk, then both strategies are the same and thus should yield equivalent payoffs, i.e.

$$e^{ni} = \frac{e^{ni^*} S}{F}$$

This is called the no-arbitrage, CIP assumption. In logs, this expression becomes

$$\frac{1}{n}(f - s) = i^* - i$$

where f and s are log values. Thus, if CIP holds, the above equality must hold, and the cross-currency basis swap spread is trivially zero.

Suppose, however, that the CIP fails to hold. This means there is a nontrivial counterparty risk. This implies there is a deviation from covered interest rate parity such that $\frac{1}{n}(f - s) \neq i^* - i$. This means that $\exists$ an $x \neq 0$, representing the deviations from CIP, such that

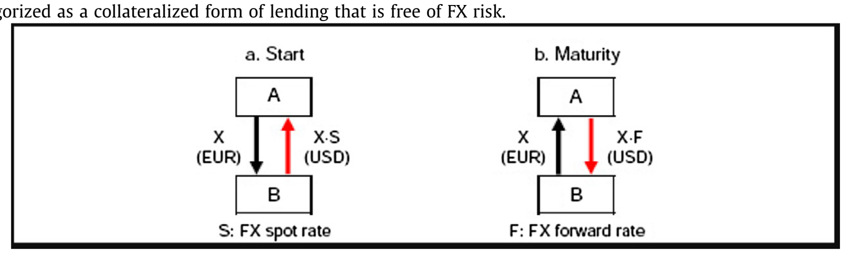
$$\frac{1}{n}(f - s) = (i^* - i) + x \tag{1.0}$$

x is known as the cross-currency basis swap spread[2].

[2] Though they serve similar economic purposes, FX swap and cross-currency basis swap contracts have different structures. Below, we provide a brief exposition on FX swap versus cross-currency basis swap contracts.

FX Swap

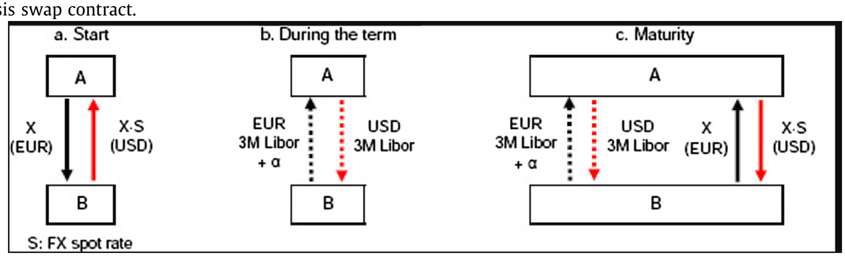
A forward exchange contract (FX swap contract in brief) is a derivative contract that market participants employ to eliminate unfavorable exposure to exchange rate vagaries. This is achieved by 'locking in' an agreed specific exchange rate (known as the forward exchange rate) to be utilized or delivered at a specified future date. Forwards are used to hedge foreign exchange risk on principal repayment. Under an FX swap contract, one party lends one currency (euro) to, and simultaneously borrows another currency (dollars) from, the second party. At maturity of contract, both parties re-exchange the principals using the agreed forward exchange rate agreed at the contract inception.This type of arrangement can be viewed as a form of collateralized lending because each party can hold on to the other party's principal as a collateral if the other party fails to pay back the borrowed principal at maturity. Thus, FX swaps are a type of secured lending and can be categorized as a collateralized form of lending that is free of FX risk.



Source: BIS Quarterly ReviewThe chart above shows the flow of funds in a euro/US dollar swap deal. Party A holds euro while party B is the US dollar entity. Party A seeks to get US dollars while party B seeks to get some euros. At inception, party A lends X euros to party B and borrows US dollars worth X·S from party B, where S is the spot exchange rate. At this inception/start of contract, an FX rate F to re-exchange the principal when the contract expires, will be agreed in advance. When the contract expires, party A receives back the X euros from party B and then pays party B the dollar value of X.F, where F is the FX forward rate agreed at inception. In FX swap contracts, only the principals are exchanged.

Cross-currency Basis Swap

While FX swap contracts involve only the exchange of principals between both parties at maturities, a cross-currency basis swap contract involves the exchange of principal at maturity as well as the exchange of interest payments periodically. The chart below illustrates the flow of funds involved in a euro/US dollar cross-currency basis swap contract.



Source: BIS Quarterly Review

At inception of a cross-currency swap contract, party A lends X euros to party B, and borrows dollars equivalent to X·S from B. During the life of the contract, party A receives floating benchmark rate of 3-month euro Libor + α from party B and pays 3-month dollar Libor to party B, where α is the price of the basis swap, a discount (positive) or premium (negative) paid to obtain the dollars, agreed upon by the two parties at inception of the contract. The exchange of the cash flows often occurs quarterly (every three months). When the contract expires, party A gets back the principal of X euros from party B and returns the dollar amount X·S to party B, where S is the same FX spot rate as of the start of the contract.

Most cross-currency basis swaps are for medium- to long-term purposes as they mirror the maturity of their underlying transactions. As such, they can range between one and 30 years in maturity whereas FX swap contracts are more short-term in nature and mostly do not extend far out into the future. Furthermore, though the structure of cross-currency basis swaps differs in some ways from FX swaps, they both serve similar economic purposes.

Meanwhile, suppose real money balances in the U.S. and eurozone is characterized as a standard function of real output and nominal interest rate, where for simplicity, we assume that the nominal interest rate is the same as the US and euro risk-free interest rates i and i^* quoted at date t in US dollars and euro. The standard Cagan-type monetary model can be written as

$$m - p = \psi_1 i + \psi_2 y \tag{1.1}$$

$$m^* - p^* = \psi_1 i^* + \psi_2 y^* \tag{1.2}$$

where m, p, i, and y denote domestic money supply, price level, interest rate and output respectively in the U.S., while asterisks denote foreign (eurozone) variables. ψ_1 and ψ_2 denote interest rate semi-elasticity of money demand and income elasticity of money demand, respectively, where $\psi_1 > 0$ and $\psi_2 > 0$ in line with Frankel (1979).

Except for interest rates, all other small-lettered variables, i.e. m, p, y, s are expressed as logarithm of money supply, price level, income, exchange rate, etc. Taking the difference between (1.1) and (1.2) gives

$$(m^* - m) - (p^* - p) = \psi_1(i^* - i) + \psi_2(y^* - y) \tag{1.3}$$

Meanwhile, standard purchasing power parity implies that, if s is the spot exchange rate, then

$$s = p^* - p, \tag{1.4}$$

We have assumed purchasing power parity holds but we do not necessarily invoke interest parity condition as extensive empirical evidence has shown that the interest parity condition is hardly supported, see Batten and Szilagyi (2011). Moreover, Macchiarelli (2011) find that while purchasing power parity holds in the long end, uncovered interest parity may not hold for some currencies.

Now, combining (1.0), (1.3), and (1.4), we obtain

$$x = \left(\frac{1}{\psi_1} - \frac{1}{n}\right)s - \frac{1}{\psi_1}(m^* - m) + \frac{\psi_2}{\psi_1}(y^* - y) + \frac{1}{n}f \tag{1.5a}$$

which links our macroeconomic variables of interest to the euro cross-currency basis swap spread, x. The stylized model provides a simple framework that leads to testable implications for the long-run relation between the euro cross-currency basis swap spreads and macroeconomic indicators – relative money supply, relative real output and spot exchange rate.

First derivative of the model with respect to our variables of interest yields

$$\frac{\partial x}{\partial s} = \left(\frac{1}{\psi_1} - \frac{1}{n}\right), \frac{\partial x}{\partial(m^* - m)} = -\frac{1}{\psi_1}, \frac{\partial x}{\partial(y^* - y)} = \frac{\psi_2}{\psi_1} \tag{1.5b}$$

Eq. (1.5a) is the model governing the long run links between the euro cross currency basis and the macroeconomic variables. The dependence of the euro basis swap spreads on the dollar spot exchange rate is in line with the mirror effect of the dollar on the basis. However, apart from Ibhagui (2020),[3] we are not aware of any other studies that link the other variables, i.e. real output and money supply dynamics, to the basis. Thus, our model simultaneously captures important potential drivers of the euro basis, some of which have not been investigated for the eurozone in the extant literature.

Assume that[4] $\psi_1 > 0$, and $\psi_2 > 1 > 0$, we have

$$\frac{\partial x}{\partial s} = \left(\frac{1}{\psi_1} - \frac{1}{n}\right) <> 0, \frac{\partial x}{\partial(m^* - m)} = -\frac{1}{\psi_1} < 0, \frac{\partial x}{\partial(y^* - y)} = \frac{\psi_2}{\psi_1} > 0, \tag{1.6}$$

implying that, ceteris paribus, an increase in relative money supply would worsen CIP deviations in the eurozone and widen the euro cross-currency basis swap, an increase in relative real output would tighten the basis while euro depreciation would either widen or tighten the cross-currency basis swaps. Also, as $\psi_2 > 1$, we have $\left|\frac{\psi_2}{\psi_1}\right| > \left|\frac{1}{\psi_1}\right| > \left|\frac{1}{\psi_1} - \frac{1}{n}\right| > 0$, suggesting that the magnitude of the impact of real output on the euro basis should be greater than the impact of relative money supply or exchange rate.

The empirical specification arising from the simple model can be written as

$$x = \beta_0 + \beta_1 s + \beta_2(m^* - m) + \beta_3(y^* - y) + \varepsilon \tag{1.7}$$

where $\beta_1 <> 0$, $\beta_2 < 0$ and $\beta_3 > 0$. Eq. (1.7) forms the framework for our empirical analysis. Note that we have excluded the last term in (1.5a) because, as our focus is on the long end of the basis, reasonably long maturity n implies $\frac{1}{n}f \to 0$.

[3] While the current article focuses on the eurozone and does not include other countries, in Ibhagui (2020) results are obtained in a panel framework and thus provide effects that are based on implicit homogeneity assumption to take advantage of pooling in panel regressions. Thus, results in Ibhagui (2020) do not necessarily extend to the specific setting studied in this paper.

[4] See Frankel (1979), Isaac and de Mel (2001), Setzer and Wolff (2009), and Nakashima and Saito (2009)

5. Data and empirical methodology

5.1. Data

The data for this study consists of quarterly observations for cross-currency basis swap spreads, nominal exchange rate (euro per U.S. dollar), eurozone broad money supply (M2) relative to U.S. broad money supply (M2), and eurozone real output relative to U.S. real output from Q2 2002 to Q3 2018, selected based on data availability for all variables. The cross-currency basis and nominal exchange rate series are from Bloomberg; the money supply and real GDP series are from Bloomberg and the Federal Reserve Economic Data (FRED). Except for cross-currency basis, all variables are measured in log-levels. Below we present a graphical view and brief commentary on the growth rates of the macroeconomic variables.

The graphs in Fig. 1c show the quarter-on-quarter growth rates of US and eurozone real output, money supply and exchange rate. Quarterly output growth has almost always been below 1% in the eurozone while in the US there are several instances where it has exceeded 1%. Moreover, in most cases, US growth has historically dominated euro-area growth. This is also reflected in the average quarterly GDP growth rate of the US which is almost twice that of the eurozone for the period. Quarterly money supply recorded high single-digit and even double-digit growth in the US and eurozone, respectively, in several quarters within the period of analysis. On average, the growth in money supply turns out higher in the eurozone than in the US. Meanwhile, the exchange rate has experienced periods of depreciation but also appreciation. The average quarterly movement in the euro exchange rate for the period under analysis appears relatively flattish. In summary, in the period of analysis, real output grew faster in US than eurozone, money supply recorded higher growth in the eurozone than US and the euro experienced periods of high appreciation and depreciation, turning out flattish on average.

5.2. Empirical methodology

We focus on the long-run relation between our variables of interest – which are the macroeconomic factors – and the long end of the euro cross-currency basis. In our time series analysis, we first provide evidence that cointegration exists for the vector $[x, s\ , (m^* - m\ ), (y^* - y\ )]$ which enables us to estimate and uncover the long-run relations for the variables, particularly how macroeconomic factors impact the euro cross-currency basis swap spreads. We estimate the cointegrating equation using appropriate time series estimators for systems of cointegrated variables. These are the dynamic ordinary least squares (DOLS), fully modified ordinary least squares (FMOLS) and canonical cointegrating regression (CCR). We also include the standard OLS for comparison.

We begin the empirical analysis by first investigating the integration properties of our series.

5.3. Unit roots

We first investigate the integration properties of $[x, s\ , (m^* - m\ ), (y^* - y\ )]$ using the well-known augmented dickey fuller (ADF) unit root test, as well as the DF-GLS and MZ_α unit root tests which have been shown to have good size and power properties, Ng and Perron (2000). Table 5 reports the results for the unit root tests. Column (1) of the first row of Table 5 shows the variable tested in levels; column 2 of the first row shows whether a linear trend was included in the unit root tests. The second row reports the first difference. All our specifications include a constant term.

We designate the variables in Table 5 as I(1) or I(0) using the following decision rule: If at least two of the tests reject the null hypothesis of nonstationarity at the 10 percent significance level or better, or if one test rejects at the 5 percent significance level, then we designate the corresponding variable I(0). If at least two of the tests do not reject the null hypothesis of nonstationarity, we designate the variable as I(1).

Based on the unit root test results in Table 5, we conclude that all four variables, $[x, s\ , (m^* - m\ ), (y^* - y\ )]$ are I(1). In this scenario, our long-run model requires cointegration between the integrated variables over our sample. Since these variables are all I(1), in the next subsection, we test for a cointegration of these four variables. We report cointegration test results for the variables in Table 6.

5.4. Cointegration test results and coefficient estimation

Table 6 reports the Johansen trace and maximum eigen value cointegration tests. The tests indicate that the null hypothesis of no cointegration is rejected, suggesting cointegrating relationship for the cross-currency basis and exchange rate, relative money supply and relative real output.

Table 7 presents the empirical results for the estimation of the coefficients of the regressions based on the specification in Eq. (1.7). Overall, several stylized features of the data, consistent with the stylized model, emerge from these regressions. First, there is a positive and strongly significant long-run relation between the cross-currency basis swap spread and relative real output in the euro area, providing support for the existence of a tightening effect of real output increases on the long-end of the euro basis. The magnitudes of the coefficients across the different estimators are remarkably similar and in line with those predicted by the benchmark model. The size of the relative real output effect on the basis is economically and statistically significant: according to the estimated point coefficients, a one percent increase in eurozone relative real output

Table 5
Unit root test results.

Variable in level	ADF	DF-GLS	MZ_α	Variable in level (with trend)	ADF	DF-GLS	MZ_α	Decision
$\boldsymbol{x}$	−0.59	−1.34	−3.63	$\boldsymbol{x}$	−1.71	−2.54	−12.91	I(1)
$\boldsymbol{s}$	−2.27	−1.59	−4.42	$\boldsymbol{s}$	−2.66	−1.94	−6.08	I(1)
$\boldsymbol{m}^* - \boldsymbol{m}$	−0.15	−0.33	−0.39	$\boldsymbol{m}^* - \boldsymbol{m}$	−1.92	−0.85	−0.88	I(1)
$\boldsymbol{y}^* - \boldsymbol{y}$	−0.91	0.96	0.63	$\boldsymbol{y}^* - \boldsymbol{y}$	−1.64	−1.61	1.29	I(1)
Variable in *first difference*	**ADF**	**DF-GLS**	MZ_α	Variable in *first difference* (with trend)	**ADF**	**DF-GLS**	MZ_α	
$\boldsymbol{x}$	−3.34***	−10.09***	−28.45***	$\boldsymbol{x}$	−3.31***	−10.09***	−28.45***	I(0)
$\boldsymbol{s}$	−7.67***	−6.11***	−28.77***	$\boldsymbol{s}$	−7.71***	−7.43***	−30.36***	I(0)
$\boldsymbol{m}^* - \boldsymbol{m}$	−2.84***	−2.86***	−8.90***	$\boldsymbol{m}^* - \boldsymbol{m}$	−3.23***	−3.17***	−10.01***	I(0)
$\boldsymbol{y}^* - \boldsymbol{y}$	−7.43***	−3.57***	−16.59***	$\boldsymbol{y}^* - \boldsymbol{y}$	−7.35***	−7.00***	−30.08***	I(0)

*, **, *** indicate significance at the 10, 5, and 1 percent levels, respectively; (trend) indicates test allows for both a constant and stationarity around a linear trend.
1. *ADF* critical values at the 10, 5, and 1 percent critical values are −2.59, −2.91 and −3.56, respectively; when a linear trend is included, the 10, 5, and 1 percent critical values become −3.17, −3.48 and −4.12, respectively.
2. *DF-GLS* critical values at 10, 5, and 1 percent levels are −1.61, −1.95 and −2.60; with linear trend included, they become −2.86, −3.15 and −3.73 respectively.
3. MZ_αcritical values at 10, 5, and 1 percent levels are −5.70, −8.10 and −13.81; with linear trend included, they become −14.20, −17.30 and –23.80 respectively.
4. H_0 in each case is the null hypothesis of nonstationarity or unit root.

Table 6
Johansen cointegration tests.

No trend					
	H_0	Trace statistic	5% Critical value	Maximum Eigen statistic	5% Critical value
	$r = 0$	57.22***	47.21	37.86***	27.58
	$r \leq 1$	28.25	29.68	19.25	21.13
	$r \leq 2$	11.64	15.41	1.69	14.26
	$r \leq 3$	1.00	3.76	2.82	3.84
With trend					
	H_0	Trace statistic	5% Critical value	Maximum Eigen statistic	5% critical value
	$r = 0$	51.14***	54.64	38.82***	32.12
	$r \leq 1$	27.86	34.55	24.92	25.82
	$r \leq 2$	15.35	18.17	10.18	19.38
	$r \leq 3$	5.05	3.74	4.32	12.52

r denotes the number of cointegrating vectors. The optimal lag lengths for the VARs were selected by minimizing the AIC. At this lag, the VAR produces results that exhibit no serial correlation. *** indicates rejection of the null hypothesis of no-cointegration at 5% level of significance or better.

Table 7
Estimation of coefficient of cointegrating equation.

	DOLS	FMOLS	CCR	OLS
$\boldsymbol{s}$	−3.22***	−1.96***	−1.95***	−1.47***
	(3.21)	(2.92)	(2.96)	(2.18)
$\boldsymbol{m}^* - \boldsymbol{m}$	−2.15***	−1.32***	−1.32***	−0.97**
	(3.05)	(2.75)	(2.77)	(1.98)
$\boldsymbol{y}^* - \boldsymbol{y}$	4.46***	3.55***	3.55***	3.43***
	(4.77)	(5.84)	(5.84)	(8.03)
$\boldsymbol{R}^2$	0.75	0.52	0.33	0.54

Figures in brackets are t-statistics. (***) and (**) indicate statistical significance at the 1% and 5% level, respectively. Test statistic in parenthesis are based on heteroskedasticity and autocorrelation corrected standard errors using Newey-West correction. For DOLS, we specify DOLS (−1, 1).

implies a long-run basis tightening of about 3.43–4.46 units. Second, the relation between cross-currency basis and relative real output is stronger than the relation between cross-currency basis and each of nominal exchange rate and relative money supply, again confirming the deduction from the stylized simple model. This can be seen to be true across all estimators employed, even for the OLS estimator which is especially the least efficient of all the estimators.

The results also show that there is a significant negative effect of euro depreciation on the cross-currency basis and this result stands for the entire estimators. The point estimate suggests that depreciation of the euro would widen the cross-currency basis by about 1.47–3.22 units in the long run. Finally, relative money supply increases are associated with basis

Table 8
Robustness with an alternative long end basis – 5-year.

	DOLS	FMOLS	CCR	OLS
$\boldsymbol{s}$	-3.99^{***}	-2.49^{***}	-2.47^{***}	-1.99^{**}
	(3.55)	(3.33)	(3.20)	(2.36)
$\boldsymbol{m^{*}-m}$	-2.85^{***}	-1.86^{***}	-1.86^{***}	-1.49^{**}
	(3.59)	(3.18)	(3.35)	(2.39)
$\boldsymbol{y^{*}-y}$	4.81^{***}	3.78^{***}	3.77^{***}	3.85^{***}
	(4.58)	(5.34)	(5.31)	(7.77)
$\boldsymbol{R}^2$	0.76	0.39	0.39	0.42

Figures in brackets are t-statistics. (***) and (**) indicate statistical significance at the 1% and 5% level, respectively. Test statistic in parenthesis are based on heteroskedasticity and autocorrelation corrected standard errors using Newey-West correction. We specific DOLS (−1, 1).

spread widening as seen in Table 7, and the relation is economically and statistically significant. A one percent rise in eurozone money supply relative to the US would widen the euro cross currency basis by about 0.97–2.15 units in the long run.

As a robustness check, we replace the 10-year basis with 5-year basis and re-estimate the model to check whether our main results are robust to the choice of the long end of the basis. The result, reported in Table 8, shows that previous findings are unaltered as all effects preserve their previous inclination and remain significant statistically and economically.

Also, the magnitude of the effect of relative real output continues to be greater than the magnitude of each of relative money supply and exchange rate respectively, suggesting that an increase in relative real output in the euro area can trigger some tightening in the long end of the euro basis which can help to lessen the widening pressure from money supply increases and or currency depreciations.

Fig. 2 presents graphs of the euro basis deviations from the level implied by our simple model for the four estimators reported in Table 7. Vertical lines are drawn for the periods Q4 2007 and Q4 2012 to roughly depict different historical regimes in the global financial markets: i) commencement of the global financial crisis and thereafter the euro crisis, and ii) the end of both the euro crisis and global financial crisis. The deviations based on each estimator have a strong tendency for mean-reversion as shown in Fig. 2. It is also shown in Fig. 2 that the deviations from the level implied by the identified macroeconomic fundamentals can be sizeable. However, except in few instances, they are generally not very persistent. Consequently, it is plausible to detect the long-run relationship between the cross-currency basis and our chosen macroeconomic variables with data inclusive of most periods represented in the data. Note that the period prior to the global financial crisis, particularly the 2004 to the 2006 period, stands out in Fig. 2 as a period where the euro cross-currency basis diverged considerably from the underlying macroeconomic fundamentals in our model. The euro cross-currency basis was little and mostly positive during this period, as is widely known in the literature. Also, the deviations of the basis from levels implied by the macro fundamentals (i.e. basis residual) tend to be quite substantial and persistent, on average, in this period and towards the end of the eurozone crisis. Because of this, it will be tough attempting to detect long-run relationships between the cross-currency basis and macroeconomic fundamentals using data from these periods alone.

To summarize the results of this section, we find considerable support for the simple stylized model of the long-run impact of macroeconomic fundamentals on the euro cross-currency basis swap spreads. Also, Fig. 2 suggests that long data spans, which include substantial periods of negligible and less persistent deviations of the basis from the identified macroeconomic factors, will generally be important to successfully detect the long-run relationship between the macroeconomic variables and the basis that is implied by our stylized model.

We now provide some views on possible routes through which the identified macroeconomic factors could influence the euro basis. The tighter basis from increased eurozone real output can be explained as follows: when eurozone relative real output improves, the eurozone becomes a more attractive investment destination. This, in turn, induces demand for euro securities. The high demand for euro assets from dollar entities – with proceeds hedged into US dollar – attracts the issuance of euro-denominated liabilities by non-euro issuers. These liabilities are subsequently swapped into US dollars. Because of the relatively benign eurozone economy, demand for eurozone securities from dollar entities would remain elevated and outstrip supply of euro securities, leading to positive net demand for euro securities (hedged into dollars) by dollar entities. This would cause the euro basis to tighten. This mechanism provides an explanation of how an increase in eurozone relative real output can tighten the euro cross-currency basis.

Turning to relative money supply, to the extent that a rise in eurozone relative money supply, which signals a more accommodative monetary stance, narrows eurozone credit spreads, this induces an increase in funding activities as foreign issuers increase their appetite for euro liabilities to latch on the lower cost of euro debt. Also, eurozone investors in search for higher US yields would take positions in US assets and hedge their dollar positions. In both cases, as the euros get swapped into US dollars, this puts a downward pressure on the euro basis, which ultimately causes it to widen. If relative money supply continues to rise, credit spreads would tighten further, issuers would remain attracted to funding via the euro, and the euro basis would widen.

Moving to the exchange rate, the result is in line with the view that dollar appreciation often goes together with wider cross-currency basis. Viewed from the perspective of supply of dollar hedging, Avdjiev et al. (2018) argues that the US dollar serves as a risk barometer, ". . .. the dollar is a key barometer of risk-taking capacity in global capital markets". Thus, a stron-

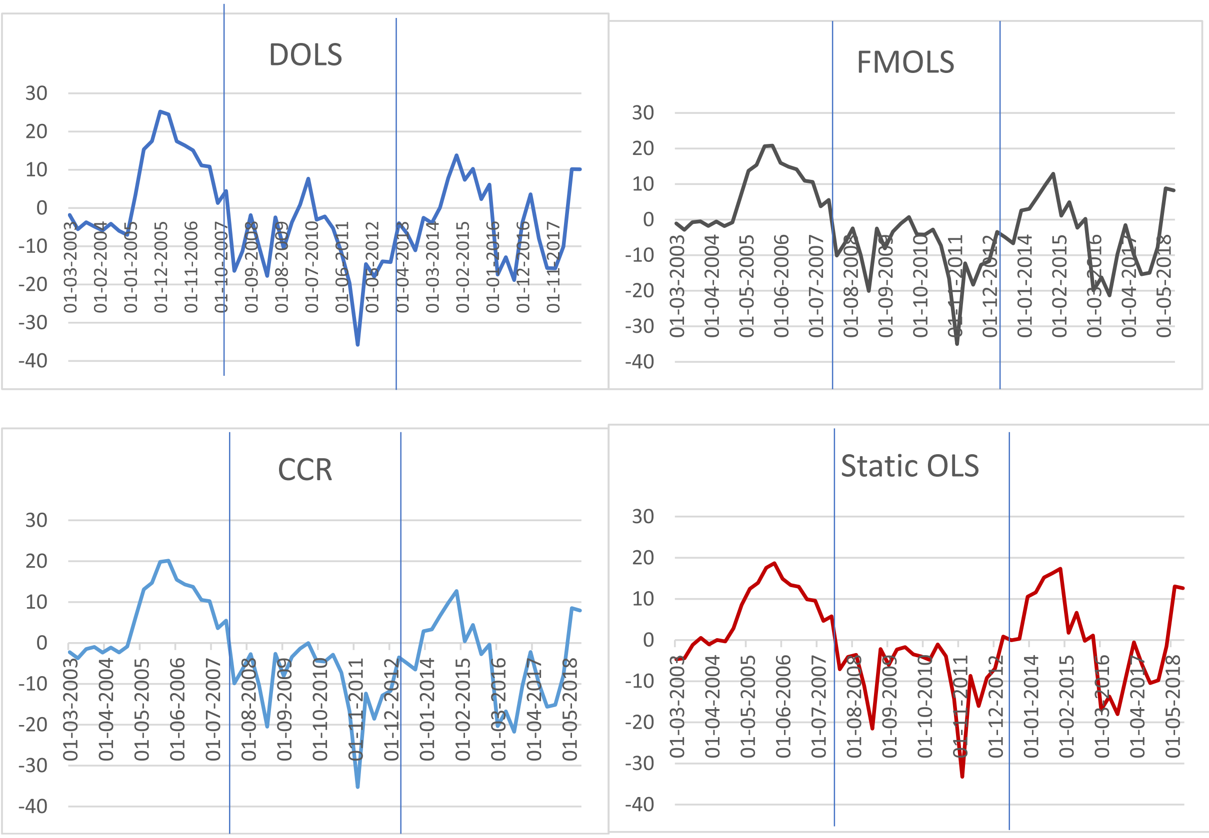


Fig. 2. Deviations of the basis from the long run level implied by our simple model. Note: Vertical lines are drawn for the periods Q4 2007 and Q4 2012 to roughly depict different historical regimes in the global financial markets: (i) commencement of the global financial crisis and thereafter the euro crisis, and (ii) the end of the euro crisis and global financial crisis.

ger dollar reduces cross-border lending in dollars as the risk of lending in dollars increases when euro weakens, or dollar strengthens. For instance, a stronger dollar means non-dollar entities with dollar liabilities become more burdened and could falter on dollar obligations. This reduces cross-border dollar lending and hence demand for euro securities, lowers supply of hedging activities and consequently widens the basis. Also, when dollar appreciates, it becomes more expensive. Thus, more euro liabilities would need to be issued and swapped to obtain the required (same) amount of dollars. The increase in issuance of euro-denominated bonds elevates demand for hedging activities, and consequently widens the euro basis. These three mechanisms provide non-exhaustive insights into how the macroeconomic variables might influence the basis.

6. Convergence to long run- error-correction mechanism

In this section, we seek to gain insight into how the long-run equilibrium is restored between the basis and the identified macroeconomic variables by estimating vector error-correction models. By the Granger representation theorem, the residual from the cointegration equation must forecast changes in at least one of the four variables x, s , $(m^* - m\ )$, $(y^* - y)$. For instance, if the residual is positive (the basis is tighter than its long-run value), either the basis must widen or one of the macroeconomic factors must shift to drive the relation back to its long-run value. Accordingly, by inspecting the error correction mechanism (ECM), insight is gained into which variables endogenously adjust to ensure convergence to the long-run relation.

Table 9 reports the dynamic impact of the error correction term on the changes of each of the variables in the cointegrated vector. We can see that for both the 5-year and 10-year basis, which are our measures for the long-end of the euro basis, it is the exchange rate and, to a much less significant extent, the euro cross-currency basis, that adjust to restore the estimated long-run equilibrium relation. In both cases, the error-correction coefficient is negative and significant while for the other variables, the error correction variable has either weak power for their dynamics or yields a positive sign unsupportive of convergence. The macroeconomic variables are thus not weakly exogeneous, and when deviations from the implied long-run equilibrium occur in the eurozone, it is predominantly the exchange rate that adjusts to restore long-run equilibrium over the sample period. In both cases, the speed of adjustment implies a half-life of about 5–6 quarters, meaning that it

Table 9
Error correction coefficients (ECM) estimate.

	(1)	(2)	(3)	(4)
		A. 10-year basis		
	x	s	$m^* - m$	$y^* - y$
ECM	−0.07	−0.14***	0.18***	0.008*
	(1.02)	(3.30)	(3.89)	(1.72)
R^2	0.16	0.15	0.20	0.17
		B. 5-year basis		
	x	s	$m^* - m$	$y^* - y$
ECM	−0.09	−0.12***	0.15***	0.010**
	(1.19)	(2.54)	(3.39)	(2.08)
R^2	0.12	0.10	0.16	0.19

Note: *,**,*** indicate significance at the 10, 5, and 1 percent levels, respectively. A. is based on the 10-year basis and B. the 5-year basis.

should take around 5–6 quarters for one half of shifts away from the equilibrium to be restored by exchange rate adjustments. Overall, the similar adjustment mechanisms at work for both tenors reflect robustness of findings for the long-end of the euro basis.

7. Conclusion

Among major financial markets, the EUR/USD cross-currency basis swap market stands out as the largest cross-currency swap market. Expectedly, there is a growing literature which so far has focused on studying the characteristics and market drivers of the EUR/USD cross-currency basis. Evidence shows that CIP deviations have expanded over time in the EUR/USD cross-currency basis swap market even across the less erratic long-end of the basis curve. However, the links between these deviations and standard macroeconomic variables remain less understood. In particular, very little has been done to assess the significance of economic factors as potential long-run drivers of the basis – that is, to assess how eurozone economic conditions influence the basis in the long run. The potential relationship between eurozone macroeconomic variables and the EUR/USD cross-currency basis at the medium to long end, which is its most active, is thus a major unanswered question in the literature since the emergence of CIP deviations. We have made attempts to address this question in this paper.

In this paper we have presented evidence supportive of a significant link between the long-end of the euro basis and eurozone macroeconomic variables. First, our result suggests that eurozone relative real output has a positive and pronounced effect on the euro basis. This means that when the eurozone real output relative to US real output increases, the euro basis tends to tighten. Second, we find that exchange rate and relative money supply also have important and economically significant effects on the euro basis in the long-run relation. Increases in relative money supply and euro depreciation both widen the euro basis. Third, we show that the magnitude of the effect of relative real output on the euro basis dominates the effect of either the exchange rate or relative money supply, so that a tightening of the basis from an increase in relative real output in the eurozone can considerably lessen the effect of a wider basis from exchange rate depreciation or money supply increases. As such, sufficiently higher real output can indeed lessen CIP deviations and lessen potential arbitrage profits in the eurozone fixed income market for dollar-based investors.

Regarding financial market policy, the existence of a long-run relation between relative real output, exchange rate, relative money supply and cross-currency basis suggests that a sustainable increase in eurozone real output position relative to the US will eventually trigger a tighter cross-currency basis, with the magnitude of tightening dominating any widening emanating from exchange rate depreciation and money supply increases. Thus, in a bid to put a floor on the widening of the basis in the long end and hence limit CIP deviations, efforts to improve real output should be favored.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.intfin.2021.101304.

References

Avdjiev, S., Du, W., Koch, C., Shin, H.S., 2018. The dollar, bank leverage and the deviation from covered interest parity. Am. Econ. Rev., Insights.
Baran, J., Witzany, J., 2017. Analyzing Cross-Currency Basis Spreads. ESM Working Paper 25–2017. European Stability Mechanism, Luxembourg.
Barnes, C., 2017. Cross Currency Swap Volumes, Clarus Financial Technology, https://www.clarusft.com/cross-currency-swap-volumes/
Batten, J.A., Szilagyi, P.G., 2011. The impact of the global financial crisis on emerging financial markets. Contemp. Stud. Econ. Financial Anal. 93, 3–16.

Cerutti, E.M., Obstfeld, M., Zhou, H., 2019. Covered interest parity deviations: Macronancial determinants. Technical report, National Bureau of Economic Research.
Correa, R., Du, W., Liao, G.Y., 2020. Us banks and global liquidity. Technical report, National Bureau of Economic Research.
Coskun and Ibhagui, 2020. CIP Deviations in Emerging Market Economies. Available at SSRN: https://ssrn.com/abstract=3623727.
Costa, M., Marçal, E.F., 2019. Deviations from covered interest parity: the role played by fundamentals, nancial and political turmoils and market frictions. Available at SSRN: https://ssrn.com/abstract=3376815
Du, W., Tepper, A., Verdelhan, A., 2018. Deviations from covered interest rate parity. J. Finance, VOL. LXXIII, NO. 3
Du, W., Hébert, B.M., Huber, A.W., 2019. Are intermediary constraints priced? Technical report, National Bureau of Economic Research.
Du, W., Im, J., Schreger, J., 2018b. The us treasury premium. J. Int. Econ. 112, 167181.
Frankel, J., 1979. On the mark: A theory of floating exchange rates based on real interest rate differentials. Am. Econ. Rev. 69 (4), 610–622.
Gargano, A., Steven, J.R., Lucio, S., 2018. The value of volume in foreign exchange, Working Paper, University of Melbourne.
Hale, G.B., Spiegel, M.M., 2012. Currency composition of international bonds: the EMU effect. J. Int. Econ. 88, 134–149.
Hartley, J.S., 2020. Covered interest rate parity deviations in external emerging market sovereign debt. J. Fixed Income 29 (4), 92–99.
Ibhagui, O., 2020. Inflation differential as a driver of cross-currency basis swap spreads. Eur. J. Finance. In press
Ibhagui, O., 2020. Covered interest parity deviations in standard monetary models. J. Econ. Business. In press
Isaac, A.G., de Mel, S., 2001. The real-interest-differential model after 20 years. J. Int. Money Finance 20, 473–495.
Jiang, Z., Krishnamurthy, A., Lustig, H., 2018. Foreign safe asset demand and the dollar exchange rate. Technical report, National Bureau of Economic Research.
Jiang, Z., Krishnamurthy, A., Lustig, H., 2020. Dollar safety and the global financial cycle. Technical report, National Bureau of Economic Research.
Liao, G., Zhang, T., 2020. The hedging channel of exchange rate determination. Available at SSRN 3612395.
Liao, G.Y., 2020. Credit migration and covered interest rate parity. J. Financial Econ. In press
Macchiarelli, C., 2011a. A VAR analysis for the uncovered interest parity and the ex-ante purchasing power parity: the role of macroeconomic and financial information. ECB working paper 1404
Nakashima, K., Saito, M., 2009. On empirical implications of highly interest elastic money demand: A note. Hitotsubashi J. Econ. 50 (1), 29–34.
Ng, S., Perron, P., 2000. Lag length selection and the construction of unit root tests with good size and power. Econometrica.
Setzer, R., Wolff, G.B., 2009. Money demand in the euro area: new insights from disaggregated data.